Supervised learning Graded Exercise

In this exercise, you will perform a data analysis using supervised learning, and write a small report. You will have to use a biomedical dataset, which can be either a private one that you have, or a public one in a public repository (check the UCI Machine Learning Repository, or Kaggle).

Aim for a concise report around 2 pages covering the main aspects of your project. Start with what the dataset is about, any acquisition details that might be relevant for the analysis. the specific problem you are solving, why it matters and why the supervised approach makes sense. Include a basic quantitative and qualitative overview of the data, followed by details on your cleaning, preprocessing, and feature engineering steps. Then, explain the models tested and the logic behind selecting them, and how the hyperparameters were set. Finally, discuss the evaluation metrics used, the train/test split and how everything was tested. Finish up with your main conclusions, and your interpretation the results.

- **Submission**
  - PDF with the report and figures,
  - Jupyter Notebook containing all outputs and charts
  - The dataset file itself, if it is public (or, if the dataset isn't public, you might be ask to do a quick demo).
  - LLM usage disclosure: a short text file detailing if/how you used any AI tools. The use of LLM tools for redacting the report is prohibited.
- **Grading breakdown:**
  - 8 points for hitting the core requirements,
  - 2 bonus points for creativity and going above and beyond.
  - The content of this report might be asked during the final exam.